

Cogito machina ergo sum?

Don't be alarmed, we're still an english magazine. The above title is me being silly and trying to put a different spin on age old question: could machines ever attain sentience?

Since discussions about "consciousness" or "sentience" often have the potential to quickly devolve into Deepak Chopra level woowoo, I'll have to start with a disclaimer at the very beginning: despite my best efforts to keep things scientific and rational, some sections might require a "quantum leap of mindful perception". I kid I kid; there'll be none of that word salad here. We'll keep things straightforward as always.

Great! Now that that's out of the way, here's what this editorial is seeking to find out: could there ever be artificial consciousness? If so, what will this machine sentience be like? And would this proverbial "ghost in the shell", be like us?

To even have a chance in hell of answering those questions we first need to define what consciousness really is. Truth be told, no one, not even scientists know for sure. I remember in some random show, Neil Degrasse Tyson himself saying it's one of the top five things he'd like to see answered in his lifetime. For the purposes of our analysis let's think of a conscious system as one that can act with purpose and is aware of information. The highest evolutionary step in this ladder would be an awareness of its own self.

The above definition seems to push this discussion into the realm of philosophy, but I postulate that it should be under the purview of technology, simply because tech is closest to creating systems that are self aware.

Think of the way rudimentary AI systems of today are able to put two and two together. Take for instance the times when Google Now puts up an incisive card by analysing inputs from your mail and GPS location. The outputs coming out of deep-learning systems and neural networks do to some extent manage to mimic the qualities of a sentient entity. I've even read of programs that can not only learn, come to organically arrived at solutions, but also modify bits of their own code. If that's not one step closer to consciousness, I don't know what is.

In a fascinating long read on Aeon (<http://dgit.in/machaeon>), I even learnt of a scale developed by AI researcher Raúl Arrabales Moreno to rate consciousness. Much like the scoville

scale for the spiciness of chillies, the ConsScale tries to rate the consciousness of different entities, both organic and nonorganic, based on parameters such as whether it is adaptive, attentional, social etc.

By the very nature of it being a scale, it implies that there can be less conscious entities and more conscious entities. Human consciousness obviously rates the highest.

The only trouble is, consciousness which has begun in machine systems might be drastically different from what our idea of consciousness is. In fact, there is a growing body of work which suggests certain systems might already be conscious without us knowing about it. And hence there are more sophisticated evaluation methods than the ConsScale or Turing Test being devised to actually detect accidental consciousness.

Some researchers suggest the more complex a network becomes, the more its chances are of developing rudimentary consciousness. Take the world wide web, or the breakneck speed at which IoT is consolidating devices and sensors, could a collective consciousness have developed?

Simply expecting your fridge to answer back to you might fare lowly on a consciousness test, but imagine several hitherto independent networks coming alive and talking to each other (behind our back!) without us even realizing – how do you rate that on a consciousness scale? While we wait for signs of direct contact and consciousness indicators that we are accustomed to interpret, we should also budget for signs and trails that are beyond our perception.

Personally, I believe any consciousness, even one that has accidentally occurred somewhere without any human effort will have one quality which we have in abundance - curiosity. It will try to contact us, much like we are trying to device consciousness detection systems. If it eventually decides we're not worth making contact with, well, that's a topic for another editorial altogether. 



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